

Jaemin Jeong

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Research Interests

Polymer-based design of the electrode–electrolyte interface in secondary batteries — how binder composition governs electrode adhesion and cycle life, and ion transport through gel and quasi-solid polymer electrolytes.

Education

Chungnam National University

B.S. in Organic Materials Engineering (transferred)

- 51 major credits completed · Cumulative GPA 3.37/4.5 · Major GPA 3.29/4.5
- Most recent semester (Spring 2026): **major GPA 3.83/4.5** over 18 credits

Daejeon, Korea
Feb 2023 – Present

Chungnam National University

B.S. in Rural and Environmental Engineering

Daejeon, Korea
Mar 2022 – Feb 2023

Republic of Korea Army

Mandatory military service

2023 – 2025

Project Experience

Fabrication and Electrochemical Characterisation of LCO Cathode Half Coin Cells

Korea Battery Academy

Team of 2 · full process carried out independently

Jul – Aug 2026

- LiCoO_2 / Super P / PVDF (94:3:3) slurry formulated and dispersed; viscosity verified at 384 s^{-1}
- Doctor-blade coating on Al foil, vacuum drying at 120°C , roll pressing, 14 mm punching — electrode thickness $81 \mu\text{m}$, loading 14.7 mg/cm^2
- Assembled CR2032 half cells against Li metal in an Ar-filled glovebox
- Formation (0.1C), C-rate (0.1–2.0C) and cycle-life testing over 3.0–4.2 V vs Li/Li^+ — 3.85 mAh discharge capacity, 97.3% initial coulombic efficiency
- **Observed up to 45% spread in initial capacity across cells from the same sheet, plus partial delamination after pressing** — the origin of my interest in binder–collector adhesion

Battery Material Characterisation (DSC · TGA · ICP-OES · GC-MS · Raman)

Korea Battery Academy

Team of 4

Jul – Aug 2026

- **DSC** — melting and crystallisation of PE/PP separator and PET; Heat–Cool–Heat programme to erase thermal history
- **TGA** — three-step decomposition of calcium oxalate monohydrate, cross-checked against theoretical stoichiometry
- **GC-MS** — 19 electrolyte solvents and additives (DMC, EMC, EC, VC, FEC) quantified in SIM mode, $R^2 > 0.999$
- **ICP-OES** — constructed a Cu calibration curve at 324.754 nm; relative standard deviation 0.395% over three replicates
- **Raman** — crystallinity and stress analysis with near-infrared excitation to suppress fluorescence

Laboratory Skills

Cell fabrication Slurry formulation and dispersion, doctor-blade coating, vacuum drying, roll pressing, punching, viscometry

Cell assembly Ar glovebox operation, CR2032 coin cell assembly

Electrochemistry Formation, C-rate, cycle life, charge–discharge profile analysis

Thermal analysis DSC (T_g / T_m / crystallisation behaviour), TGA (decomposition, thermal stability)

Composition ICP-OES, GC-MS, Raman, IC/MS

Computing Python (pandas, matplotlib) — experimental data handling

Awards and Certification

Secondary Battery Material Analysis and Recycling — 132-hour course completed

Korea Battery Academy

Korea Battery Industry Association (KBIA) · Certificate No. 2026-00048

Jul – Aug 2026

Daegu Public Data Startup Competition — Award

Daegu Digital Innovation Promotion Agency

Idea and Planning Division · public data collection and preprocessing

2025

Relevant Coursework

Polymer Synthesis (A0) · Polymer Materials (A0) · Industrial Chemistry I (A+) · Engineering Physics (A0)

Organic Material Properties · Organic Materials · Introduction to Polymer Engineering · Physical Chemistry and Laboratory · Introduction to Materials Engineering · Textile Chemistry · Petrochemical Industry

Languages: Korean (native), English